ZJU-UIUC INSTITUTE
Zhejiang University-University of Illinois Urbana-Champaign Institute
浙江大学伊利诺伊大学
厄巴纳香槟校区联合学院

# Curriculum of Computer Engineering For Class of 2028

**Zhejiang University/University of Illinois Urbana-Champaign Institute**

**(ZJU-UIUC INSTITUTE)**

# Table of Contents

1. Overview ..... 2
2. Graduation Requirement ..... 2
3. Required Courses ..... 4
4. Elective Courses ..... 6
5. ZJU Required General Education ..... 11
6. Course Schedule Sample by Semester ..... 12
   6.1 First Year-First (Fall) Semester ..... 12
   6.2 First Year-Second (Spring) Semester ..... 13
   6.3 Second Year-First (Fall) Semester ..... 14
   6.4 Second Year-Second (Spring) Semester ..... 14
   6.5 Third Year-First (Fall) Semester ..... 15
   6.6 Third Year- Second (Spring) Semester ..... 15
   6.7 Fourth Year-First (Fall) Semester ..... 15
   6.8 Fourth Year- Second (Spring) Semester ..... 16
7. Curriculum Flow Map ..... 18

# 1. Overview

The Zhejiang University/University of Illinois Urbana-Champaign Institute (ZJU-UIUC Institute, ZJUI) offers a joint dual-degree engineering program. Upon successful completion of the program, and after meeting the graduation requirements of both universities, students will obtain bachelor's degrees separately from Zhejiang University (ZJU) and University of Illinois Urbana-Champaign (UIUC).

Computer engineering focuses on the development of vital computing technologies, ranging from chips to computers to networks to programming tools to key algorithms for building exciting applications. Fundamentally, computer engineering addresses the problem of building scalable, trustworthy computing systems, and the faculty's interests span a broad spectrum of issues pertinent to this theme. Computer engineering has taken the lead in revolutionizing many science and engineering disciplines with parallel computing, from chips to clouds to planet-scale critical infrastructures, and has defined new standards of security, privacy, and dependability for systems ranging from small circuits to the electric power grids of many nations. Students need a broad and sound set of mathematical and computing skills and are well-served by a flexible curriculum that enables them to pursue topics of interest among the many sub-disciplines in computing.

The core curriculum of the Computer Engineering program focuses on the fundamental computer engineering knowledge: circuits, systems, electromagnetics, computer systems, electronics for information processing and communication, and computer science. The rich set of ECE elective courses permits students to concentrate on any sub-discipline of computer engineering, including hardware systems; cyber physical systems; foundations and theory; software and languages; algorithms and mathematical tools; trust, reliability, security; networking, mobile and distributed computing; big data analytics and systems; artificial intelligence, robotics, cybernetics.

During their junior and senior years students are encouraged to take courses and participate in research projects in interdisciplinary areas with Civil and Environmental Engineering, Mechanical Engineering, Electrical Engineering, and others.

# 2. Graduation Requirement

## 2.1 Graduation Requirements for UIUC degree

### 2.1.1 Grade Point Average Requirement

A student must maintain a minimum GPA of 2.0 (A=4.0) to remain in good standing and graduate. UIUC degree is subjected to the final review by UIUC.

### 2.1.2 Junior Eligibility Requirement

To be eligible to enroll in the ECE courses listed in the third year of the curriculum, a student must have completed, with a combined 2.25 grade point average, the mathematics, physics, computer science, and electrical and computer engineering courses listed in the first two years.

### 2.1.3 Curriculum Requirement

The curriculum leading to the degree of Bachelor of Science in Computer Engineering at UIUC requires **128** academic credits (or academic units, AUs) and is organized into required courses, technical electives, general education, and other electives.

| **i). Required courses, see section 3 for details.** | Academic Credits |
|---|---|
| a) Orientation and Professional Development | 1 |
| b) Basic Sciences and Mathematics | 29 |
| c) Technical Core | 36 |
| d) Composition | 8 |
| e) Advanced Composition* | 4 |
| **Total Required AUs:** | **78** |

| **ii). Elective courses, see section 4 for details.** | Academic Credits |
|---|---|
| a) Technical Electives | 30 |
| b) General Education | 12 |
| c) Free Electives | 12 |
| **Total Required AUs:** | **54** |

**Students take ECE 445 to satisfy advanced composition and technical elective requirement, and credits can be given for both.*

### 2.1.4 Additional requirements

For UIUC degree, in addition to the specific course and grade point average requirements listed above, each candidate for a bachelor's degree from UIUC must meet the following requirements:

**i). Residency Requirement**

Earn a minimum sixty (**60**) academic credits of UIUC credit, of which at least **twenty-one** (**21**) AUs must be 300 or 400 level UIUC credit courses.

**ii). Transfer Requirement**

Students should have a satisfactory English Proficiency Test score on TOEFL, IELTS or other exams approved by UIUC, and maintain a good standing on academic studies that all term GPAs, overall GPA on UIUC courses are suggested to be above 2.5. In junior year, students will apply for international transfer to be admitted to UIUC and change academic status from non-degree student to degree student in Fall semester of senior year.

## 2.2 Graduation Requirements for ZJU degree

In addition to the 128-AU requirement listed above, the curriculum leading to the degree of Bachelor of Engineering from ZJU requires students to complete additional **33.5** academic credits of ZJU-required general education courses for domestic students and additional seven courses (**23** AUs) in Chinese language and society study for international students. Please refer to section 5 for details.

# 3. Required Courses

## 3.1 Orientation and Professional Development

This course introduces students to the opportunities and resources our institute and curriculum can offer you as you work to achieve your career goals. It also provides the skills to work effectively and successfully in the engineering profession.

| Course Code | Course Name | Academic Credit |
|---|---|---|
| ENG 100 | Engineering Orientation | 1 |
| | **Total Required AUs:** | **1** |

## 3.2 Foundational Mathematics and Science

These courses emphasize the fundamental mathematical and scientific principles upon which the engineering discipline is based.

| Course Code | Course Name | Academic Credits |
|---|---|---|
| MATH 221 | Calculus I | 4 |
| MATH 231 | Calculus II | 3 |
| MATH 241 | Calculus III | 4 |
| MATH 257 | Linear Algebra with Computational Application | 3 |
| MATH 285 | Introduction to Differential Equation | 3 |
| PHYS 211 | University Physics: Mechanics | 4 |
| PHYS 212 | University Physics: Elec & Mag | 4 |
| PHYS 213 | University Physics: Thermal Physics | 2 |
| PHYS 214 | University Physics: Quantum Physics | 2 |
| | **Total Required AUs:** | **29** |

## 3.3 Technical Core

These courses emphasize fundamental concepts and basic laboratory techniques that comprise the common intellectual understanding of electrical engineering.

| Course Code | Course Name | Academic Credits |
|---|---|---|
| ECE 110 | Introduction to Electronics | 3 |
| ECE 120 | Introduction to Computing | 4 |
| ECE 210 | Analog Signal Processing | 4 |
| ECE 220 | Computer Systems & Programming | 4 |
| CS 173* | Discrete Structures | 3 |
| CS 225 | Data Structures | 4 |
| ECE 313** | Probability with Engineering Application | 3 |
| ECE 374 | Introduction to Algorithms & Models of Computation | 4 |
| ECE 385 | Digital Systems Laboratory | 3 |
| ECE 391 | Computer Systems Engineering | 4 |
| | **Total Required AUs:** | **36** |

** CS 173 can be substituted by Math 213 (Basic Discrete Mathematics, 3 AUs).*
*** ECE 313 can be substituted by STAT 410 (Statistics and Probability II, 3 AUs)*

## 3.4 Composition

i) These courses teach fundamentals of expository writing.

| Course Code | Course Name | Academic Credits |
|---|---|---|
| RHET 101 | Principles of Writing | 4 |
| RHET 102 | Principles of Research | 4 |
| | **Total Required AUs:** | **8** |

## 3.5 Advanced Composition

The Advanced Composition requirement is fulfilled by a writing-intensive course beyond basic composition. It is normally taken in the junior or senior year.

| Course Code | Course Name | Academic Credits |
|---|---|---|
| ECE 445* | Senior Design Project Lab | 4 |
| | **Total Required AUs:** | **4** |

** ECE 445 is also a required technical elective.*

# 4. Elective Courses

## 4.1 Technical Electives

These courses stress cross-disciplinary study, the rigorous analysis and design principles practiced in the major sub-disciplines of computer engineering, for example cyber physical systems; foundations and theory; software and languages; algorithms and mathematical tools; trust, reliability, security; networking, mobile and distributed computing; big data analytics and systems; artificial intelligence, robotics, cybernetics, etc.

Students must complete a minimum of **thirty (30)** academic credits in technical electives, including at least:

| Number of course | Types of elective courses | Course list | Academic Credits |
|---|---|---|---|
| 1) 1 | **EE Foundations** courses | section 4.1.1 | |
| 2) 3 | **Advanced Computing Elective** courses | section 4.1.2 | |
| 3) 1 | **ECE 445** - Senior Design Project Lab | | |
| 4) | Other technical elective courses | section 4.1.4 | |
| | | **Total required AUs:** | **30** |

For courses that fall in multiple categories of technical electives, the academic units of the course will be counted to satisfy the AU requirements of each category, but will be counted only once in satisfying the total minimum academic unit requirements (30 AUs) of technical electives.

### 4.1.1 EE Foundation Courses

Select **One** (**1**) course from the following 6 EE Foundation Courses:

**ECE**: 310, 329, 330, 340, 461, 486.

| Course Code | Course Name | Academic Credits |
|---|---|---|
| ECE 310 | Digital Signal Processing | 3 |
| ECE 329 | Fields and Waves I | 3 |
| ECE 330 | Power Circuits & Electromechanics | 3 |
| ECE 340 | Semiconductor Electronics | 3 |
| ECE 461 | Digital Communications | 3 |
| ECE 486 | Control Systems | 4 |

### 4.1.2 Advanced Computing Elective Courses

Following CS and ECE courses are listed as advanced computing electives. Some ECE ones are cross listed with CS.

Select **Three** (**3**) courses from the courses listed in the following table:

| Course Code | Course Name | Academic Credits |
|---|---|---|
| CS 357 | Numerical Methods I | 3 |
| CS 411 | Database Systems | 3 |
| CS 412 | Introduction to Data Mining | 3 |
| CS 414 | Multimedia Systems | 3 |
| CS 418 | Interactive Computer Graphics | 3 |
| CS 419 | Production Computer Graphics | 3 |
| CS 420 | Parallel Programming: Scientists and Engineers | 3 |
| CS 421 | Programming Languages & Compilers | 3 |
| CS 423 | Operating Systems Design | 3 |
| CS 424 | Real-Time Systems | 3 |
| CS 425 | Distributed Systems | 3 |
| CS 426 | Compiler Construction | 3 |
| CS 431 | Embedded Systems | 3 |
| CS 436 | Computer Networking Laboratory | 3 |
| CS 437 | Topics in Internet of Things | 3 |
| CS 438 | Communication Networks | 3 |
| CS 440 | Artificial Intelligence | 3 |
| CS 441 | Applied Machine Learning | 3 |
| CS 444 | Deep Learning for Computer Vision | 3 |
| CS 446 | Machine Learning | 3 |
| CS 450 | Numerical Analysis | 3 |
| CS 461 | Computer Security I | 3 |
| CS 473 | Algorithms | 4 |
| CS 475 | Formal Models of Computation | 3 |
| CS 476 | Program Verification | 3 |
| CS 477 | Formal Software Development Methods | 3 |
| CS 483 | Applied Parallel Programming | 3 |
| ECE 407/CS 407 | Cryptography | 3 |

| | | |
|---|---|---|
| ECE 408/CS 483 | Applied Parallel Programming | 4 |
| ECE 411 | Computer Organization & Design | 4 |
| ECE 412 | Microcomputer Laboratory | 3 |
| ECE 419/CS 460 | Computer Security Laboratory | 3 |
| ECE 422/CS 461 | Computer Security I | 4 |
| ECE 424/CS 463 | Computer Security II | 3 |
| ECE 425 | Intro to VLSI System Design | 3 |
| ECE 428/CS 425 | Distributed Systems | 3 |
| ECE 434 | Mobil Computing and Application | 3 |
| ECE 435/CS 436 | Computer Networking Laboratory | 3 |
| ECE 438/CS 438 | Communication Networks | 3 |
| ECE 439/CS 439 | Wireless Networks | 3 |
| ECE 448/CS 440 | Artificial Intelligence | 3 |
| ECE 449/CS 446 | Machine Learning | 3 |
| ECE 462 | Logic Synthesis | 3 |
| ECE 470 | Introduction to Robotics | 4 |
| ECE 478/CS 477 | Formal Software Development Methods | 3 |
| ECE 479 | LoT and Cognitive Computing | 3 |
| ECE 484 | Principles of Safe Autonomy | 4 |
| ECE 491/CS 450 | Numerical Analysis | 3 |
| ECE 492/CS 420 | Parallel Programming: Scientists and Engineers | 3 |

### 4.1.3 Other Electives

Students are encouraged to build up their cross-discipline study by taking non-ECE technical courses from Civil Engineering, Mechanical Engineering, Computer Science, and others at ZJUI or during exchange at UIUC.

Select sufficient technical electives to satisfy the minimum academic units of technical electives (30 AUs).

Note: The academic units of the courses in 4.1.1 and 4.1.2 will not be counted in 4.1.3.

**Aerospace Eng. (AE):** 202, 302, 311, 312, 321, 352, 353, 402, 403, 410, 412, 416, 419, 420, 428, 433, 434, 435, 451, 460

**Agri. Bio Eng. (ABE):** any 300- and 400-level course[2] EXCEPT 440

**Astronomy (ASTR):** 210, 310, 330, 350, 404, 405, 406, 414, 450

**Atmospheric Science (ATMS):** 201, 301, 302, 303, 304, 305, 404, 405, 406, 410, 411, 420, 421, 447, 449

**Biochemistry (BIOC):** 406, 440, 446, 455

**Bioengineering (BIOE):** 201, 202, 302, 414, 415, 461, 467, 473, 476, 480, 485

**Biophysics (BIOP):** all 400-level courses[1]

**Chem & Bio Eng (CHBE):** 221, 321, 421, 422, 424, 430, 431, 440, 451, 452, 453, 456, 457, 471, 472, 473, 474

**Chemistry (CHEM):** Chem 104, 105, any 200/300/400-level[2] class EXCEPT 396/7, 497, 499

**Civil & Env. Eng. (CEE):** 310, 330, 408, 410, 416, 430, 447, 491

**Crop Sciences (CPSC)**: 265

**Computer Science (CS)**: 242, 357, 410, 411, 412, 413, 414, 415, 416, 418, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429, 431, 433, 436, 438, 439 ,440, 445, 446, 447, 450, 460, 461, 463, 465, 466, 467, 473, 475, 476; 477, 481, 484.

**ECE:** 304, 307, 310, 311, 314, 329, 330, 333, 340, 342, 343, 350, 365, 380, 395, 402, 403, 407, 408, 410, 411, 412, 414, 415, 416, 417, 418, 419, 420, 421, 422, 424, 425, 428, 431, 432, 434, 435, 437, 438, 439, 441, 443, 444, 445, 446, 447, 448, 449, 451, 452, 453, 454, 455, 456, 457, 458, 459, 461, 460, 462, 463, 464, 465, 466, 467, 468, 469, 470, 471, 472, 473, 476, 478, 479, 480, 481, 482, 483, 484, 485, 486, 487, 488, 489, 490, 491, 492, 493, 495.

**Geology (GEOL):** 107, 208, 333, 380, 411, 417, 432, 436, 440, 450, 452, 460

**Industrial Eng. (IE):** 310, 330, 360, 361, 400, 410, 411, 412, 413, 420, 430, 431

**Integrative Biology (IB):** 150, 202, 203, 204, 302, 335, 348, 368, 401, 405, 420, 421, 426, 427, 431, 432, 440, 443, 444, 451, 452, 453, 461, 462, 463, 464, 467, 468, 471, 472, 473, 481, 482, 483, 485, 486

**Linguistics (LING):** 300, 406, 407, 427

**Material Science & Engineering (MSE):** 280, any 300/400-level[1] class EXCEPT 304, 460, 461

**Math:** 347, 348, 357, 402, 403, 412, 413, 414, 415, 416, 417, 418, 423, 424, 425, 427,428, 432, 442, 444, 446, 447, 448, 450, 453, 473, 475, 481, 482, 484, 487, 489, 494

**Molecular & Cellular Biology (MCB):** 150, 250, 251, 252, 253, 300, 301, 314, 316, 354, 400, 401, 402, 403, 404, 406, 408, 410, 413, 419, 421, 424, 426, 430, 431, 433, 435, 446, 480

**Mechanical Eng. (ME):** 200, 310, 320, 330, 340, 370, 371, 400, 401, 402, 403, 404, 410, 411, 412, 420, 430, 431, 440, 445, 451, 452, 460, 461, 471, 472, 485, and 487

**Music (MUS):** 407, 409

**Neuroscience (NEUR):** 453

**Nuclear Plasma & Radiological (NPRE):** 201, 247, 402, 412, 421, 423, 429, 431, 432, 435, 441, 442, 444, 446, 447, 448, 451, 455, 457, 458, 470, 475

**Physics (PHYS):** 225, 246, 325, 326, 370, 371, 401, 402, 403, 406, 419, 420, 427, 460, 466, 470, 485, 486, 487

**Psychology (PSYC):** 204

**Speech & Hearing Science (SHS):** 200, 240, 300, 301, 320, 450, and 470

**Statistics (STAT):** 420, 424, 425, 428, 429, and 440

**Systems Eng (SE):** 411, 420, 423, 424

**Theoretical & Applied Mechanics (TAM):** 211, 212, 251, 324, 335, 412, 435, 445, 451

[1] except seminars and special topics courses, which may be reviewed in the Advising Office.

## 4.2 General Education

The general education courses develop students' understanding of human culture and society, build skills of inquiry and critical thinking, and lay a foundation for civic engagement and lifelong learning. To satisfy the General Education requirements, students must complete a minimum of **twelve (12)** AUs in general education courses. Students must take all courses for grade and complete sufficient courses in required categories based on the table below:

| Required course category | Number of courses | Academic Credits |
|---|---|---|
| 1) **Humanities & Arts** | Two (2) | 6 |
| 2) **Social & Behavioral Sciences**[1] | Two (2) | 6 |
| Additional Requirements: | | |
| **i)** Culture Studies[2] | | |
| Western/Comparative Culture(s) | One (1) | 3 |
| Non-Western Culture(s) | One (1) | 3 |
| U.S. Minority Culture(s)[3] | One (1) | 3 |
| **ii)** Aesthetic Education[4] | One (1) | 3 |
| | **Total required AUs:** | 12 |

[1] One of the Social and Behavioral Science courses is recommended to be an introductory economics course (ECON 102 or ECON 103).
[2] Proper selection of Social and Behavioral Sciences and in Humanities and the Arts courses will assure that these courses also fulfill Western, non-Western and US Minority Cultures requirements.
[3] Not required if students pursue ZJU degree only.
[4] Required only for ZJU degree.

For courses that fall in multiple categories of general education, the academic units of the course will be counted to satisfy the AU requirements of each category, but be counted only once in satisfying the total minimum academic unit requirements (12 AUs) of general education.

### 4.3 Free Electives

These unrestricted electives give students the opportunity to explore any intellectual area of unique interest. This freedom plays a critical role in helping students define their research specialties. Students are encouraged to take cross-disciplinary courses as free electives.

Students must complete a minimum of **twelve (12)** AUs in free electives.

## 5. ZJU Required General Education

These courses introduce modern Chinese history, social development, government policies, etc., and help students improve their English and maintain a healthy lifestyle.

### 5.1 For domestic students

Domestic students must complete all the courses below to fulfill the graduation requirements for the Bachelor of Engineering Degree at ZJU. These courses can be taught in Chinese. The academic credits in this section are additional to 128 AUs of credits required for UIUC degree.

| Course Code | Course Name | Academic Credits |
|---|---|---|
| LAW 1002 | Ideology, Morality and Rule of Law | 3 |
| HIST 2001 | Modern Chinese History | 3 |
| PHIL 2002 | Introduction to the Principle of Marxism | 3 |
| PS 2011 | Intro.to Mao Thought & Theoretical System of China Socialism | 3 |
| PS 3011 | General Introduction to Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era | 3 |
| PS 1001 | Situation and Policy I | 1 |
| PS 2001 | Situation and Policy II | 1 |
| ENGL 1001 | Integrated English I | 3 |
| ENGL 1002 | Integrated English II | 3 |
| PE 1001 | Physical Education I | 1 |
| PE 1002 | Physical Education II | 1 |
| PE 2001 | Physical Education III | 1 |
| PE 2002 | Physical Education IV | 1 |
| PE 4021 | Physical Education VII-Fitness test and exercise | 0.5 |
| MITR 1001 | Military Training | 2 |
| MITR 2001 | Military Theory | 2 |

| Course Code | Course Name | Academic Credits |
|---|---|---|
| ST 4001 | Senior Thesis | 2 |
| | **Total Required AUs:** | **33.5** |

### 5.2 For International students

International students are required to complete the following courses in Chinese language study and Chinese society to fulfill the graduation requirements along with the above 128 academic credits of courses towards the Bachelor of Engineering Degree at ZJU.

| Course Code | Course Name | Academic Credits |
|---|---|---|
| CHIN 1001 | Chinese I | 4 |
| CHIN 1002 | Chinese II | 4 |
| CHIN 1003 | Chinese III | 4 |
| CHIN 1004 | Chinese IV | 4 |
| CHIN 1005 | International Chinese Language Proficiency and Testing | 2 |
| CULT 2001 | China Survey | 3 |
| ST 4001 | Senior Thesis | 2 |
| | **Total Required AUs:** | **23** |

## 6. Course Schedule Sample by Semester

### 6.1 First Year-First (Fall) Semester

List of ZJUI/UIUC courses for both domestic and international students

| No | Course Code | Course Name | Academic Credits |
|---|---|---|---|
| 1 | Rhet 101 | Principles of Writing | 4 |
| 2 | Math 221* | Calculus I | 4 |
| 3 | ECE 110 | Intro to Electronics | 3 |
| 4 | ECE 120 | Introduction to Computing | 4 |
| 5 | Eng 100 | Engineering Orientation | 1 |
| 5 | GenEd 1 | | 3 |
| | | Total: | 19 |

** MATH 213 can be selected if MATH 221 is exempted.*

List of ZJU courses for domestic students

| No | Course Code | Course Name | Academic Credits |
|---|---|---|---|
| 1 | MITR 1001 | Military Training | 2 |
| 2 | ENGL 1001 | Integrated English I | 3 |
| 3 | PHIL 2002 | Introduction to the Principle of Marxism | 3 |
| 4 | PE 1001 | Physical Education I | 1 |
| 5 | PS 1001 | Situation and Policy I | |

List of ZJU courses for international students

| No | Course Code | Course Name | Academic Credits |
|---|---|---|---|
| 1 | CHIN 1001 | Chinese I | 4 |

## 6.2 First Year-Second (Spring) Semester

List of ZJUI/UIUC courses for both domestic and international students

| No | Course Code | Course Name | Academic Credits |
|---|---|---|---|
| 1 | Rhet 102 | Principles of Research | 4 |
| 2 | Math 231 | Calculus II | 3 |
| 3 | Phys 211 | University Physics: Mechanics | 4 |
| 4 | ECE 220 | Computer Systems & Programming | 4 |
| 5 | MATH 257 | Linear Algebra with Computational Application | 3 |
| | | Total: | 18 |

List of ZJU Courses for domestic students

| No | Course Code | Course Name | Academic Credits |
|---|---|---|---|
| 1 | LAW 1002 | Ideology, Morality and Rule of Law | 3 |
| 2 | ENGL1002 | Integrated English II | 3 |
| 3 | PE1002 | Physical Education II | 1 |
| 4 | PS 1001B | Situation and Policy I | 1 |

List of ZJU Courses for international students

| No | Course Code | Course Name | Academic Credits |
|---|---|---|---|
| 1 | CHIN 1002 | Chinese II | 4 |

## 6.3 Second Year-First (Fall) Semester

List of ZJUI/UIUC courses for both domestic and international students

| No | Course Code | Course Name | Academic Credits |
|---|---|---|---|
| 1 | Math 241 | Calculus III | 4 |
| 2 | Phys 212 | University Physics: Elec& Mag | 4 |
| 3 | ECE 313/314 | Probability with Engineering Applications/Lab | 4 |
| 4 | Math 213 | Basic Discrete Mathematics | 3 |
| 5 | GenEd 2* | General Education Elective | 3 |
| | | Total: | 18 |

* Suggest selecting ECON 102 or ECON 103.

List of ZJU Courses for domestic students

| No | Course Code | Course Name | Academic Credits |
|---|---|---|---|
| 1 | PS 2011 | Intro.to Mao Thought & Theoretical System of China Socialism | 5 |
| 2 | PE 2001 | Physical Education III | 1 |
| 3 | MITR 2001 | Military Theory | 2 |
| 4 | PS 2001 | Situation and Policy II | |

List of ZJU Courses for international students

| No | Course Code | Course Name | Academic Credits |
|---|---|---|---|
| 6 | CHIN 1003 | Chinese III | 4 |
| 7 | CULT 2001 | China Survey | 3 |

## 6.4 Second Year-Second (Spring) Semester

List of ZJUI/UIUC courses for both domestic and international students

| No | Course Code | Course Name | Academic Credits |
|---|---|---|---|
| 1 | Math 285 | Intro to Differential Eq | 3 |
| 2 | Phys 213 | Univ Physics: Thermal Physics | 2 |
| 3 | Phys 214 | Univ Physics: Quantum Physics | 2 |
| 4 | ECE 210 | Analog Signal Processing | 4 |

| | | | |
|---|---|---|---|
| 5 | CS 225 | Data Structure | 4 |
| 6 | GenEd 3 | General Education Elective | 3 |
| | | Total: | 18 |

List of ZJU Courses for domestic students

| No | Course Code | Course Name | Academic Credits |
|---|---|---|---|
| 1 | HIST 2001 | Modern Chinese History | 3 |
| 2 | PS 2001 | Situation and Policy II | |
| 3 | PE 2002 | Physical Education IV | 1 |

List of ZJU Courses for international students

| No | Course Code | Course Name | Academic Credits |
|---|---|---|---|
| 1 | CHIN 1004 | Chinese IV | 4 |

## 6.5 Third Year-First (Fall) Semester

| No | Course Code | Course Name | Academic Credits |
|---|---|---|---|
| 1 | EE Foundation | One of EE Foundations courses | 3 |
| 2 | ECE 391 | Computer Systems Engineering | 4 |
| 3 | Tech Elec | Technical Elective | 3 |
| 4 | Tech Elec | Technical Elective | 3 |
| 5 | GenEd 4 | General Education Elective | 3 |
| | | Total: | 16 |

## 6.6 Third Year- Second (Spring) Semester

| No | Course Code | Course Name | Academic Credits |
|---|---|---|---|
| 1 | ECE 374 | Introduction to Algorithms & Models of Computation | 4 |
| 2 | ECE 385 | Digital Systems Laboratory | 3 |
| 3 | Tech Elec | Technical Elective | 3 |
| 4 | Tech Elec | Technical Elective | 3 |
| | | Total: | 13 |

## 6.7 Fourth Year-First (Fall) Semester

List of ZJUI/UIUC courses for both domestic and international students

| No | Course Code | Course Name | Academic Credits |
|---|---|---|---|
| 1 | Tech Elec | Technical Elective | 3 |
| 2 | Tech Elec | Technical Elective | 3 |
| 3 | Tech Elec | Technical Elective | 3 |
| 4 | Free Elec | Free Elective | 3 |
| 5 | Free Elec | Free Elective | 3 |
| | | Total: | 15 |

List of ZJU Courses for domestic students

| No | Course Code | Course Name | Academic Credits |
|---|---|---|---|
| 1 | PS 3011 | General Introduction to Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era | 3 |
| 2 | PS 2001 | Situation and Policy II | |

## 6.8 Fourth Year- Second (Spring) Semester

List of ZJUI/UIUC courses for both domestic and international students

| No | Course Code | Course Name | Academic Credits |
|---|---|---|---|
| 1 | ECE 445 | Senior Design Project Lab | 4 |
| 2 | Tech Elec | Technical Elective | 3 |
| 3 | Tech Elec | Technical Elective | 3 |
| 4 | Free Elec | Free Elective | 3 |
| | | Total: | 13 |

List of ZJU Courses for domestic students

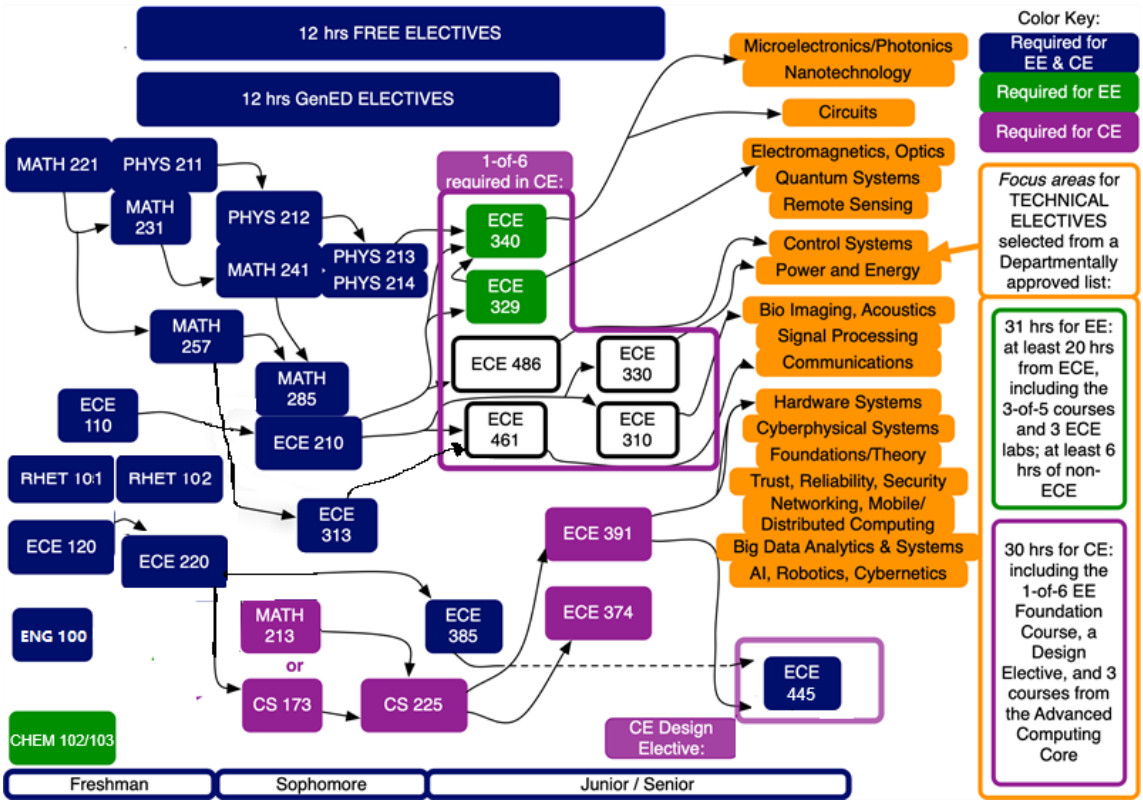
| No | Course Code | Course Name | Academic Credits |
|---|---|---|---|
| 1 | PE 4021 | Physical Education VII-Fitness test and exercise | 0.5 |
| 2 | PS 2001F | Situation and Policy II | 1 |
| 3 | ST 4001 | Senior Thesis | 2 |

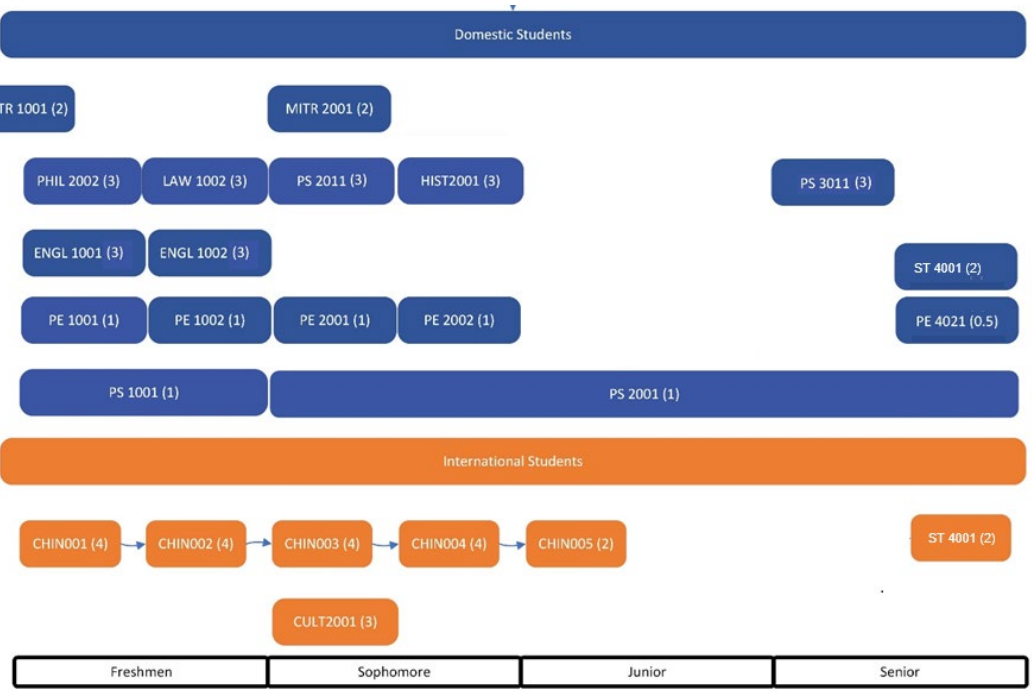
List of ZJU Courses for international students

| No | Course Code | Course Name | Academic Credits |
|---|---|---|---|
| 1 | ST 4001 | Senior Thesis | 2 |

# 7. Curriculum Flow Map

The following flow map provides a quick summary of the main features of the Computer Engineering curriculum.




ZJU Required General Education Course